

Mathematical pendulum in the field of a ball

X16 (2016) — English translation

A mathematical pendulum makes small oscillations near the surface of a dielectric ball uniformly charged with negative electricity. The length of the pendulum is equal to the radius of the ball.

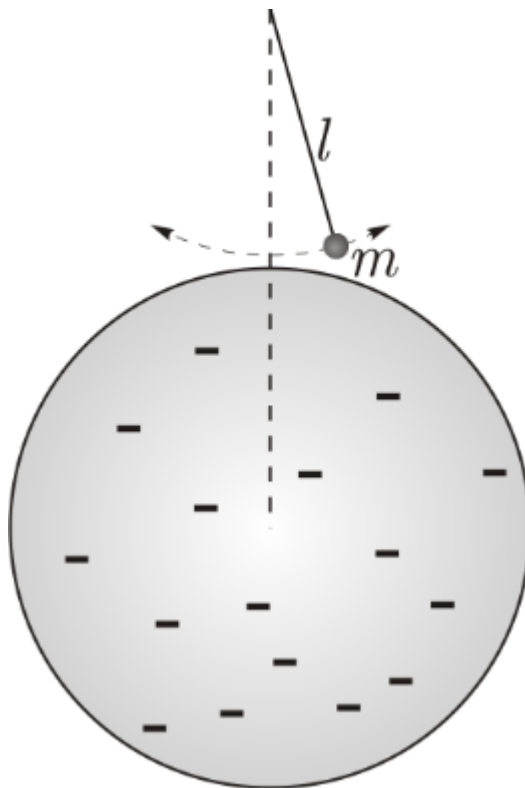


Fig. 1.

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| A1 | What is the period of harmonic oscillations of a pendulum if the length of its thread is l , the mass of the weight is m , the charge is $q > 0$, and the magnitude of the field strength near the surface of the ball is E_0 . | 2.00 |
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Ignore gravity and the field of induction charges

Source: <https://pho.rs/p/3998>